Rewarding Learning

ADVANCED SUBSIDIARY (AS)
General Certificate of Education
2019

Centre Number

Candidate Number

Digital Technology

Assessment Unit AS 2
assessing
Fundamentals of Digital Technology

SDT21

[SDT21]
FRIDAY 24 MAY, AFTERNOON

TIME

1 hour 30 minutes.

INSTRUCTIONS TO CANDIDATES

Write your Centre Number and Candidate Number in the spaces provided at the top of this page.
You must answer the questions in the spaces provided.
Do not write outside the boxed area on each page or on blank pages.
You are not permitted to use any calculating aid in this paper.
Complete in black ink only. **Do not write with a gel pen.**
Answer **all five** questions.

INFORMATION FOR CANDIDATES

The total mark for this paper is 100.
Figures in brackets printed down the right-hand side of pages indicate the marks awarded to each question or part question.
Quality of written communication will be assessed in Questions **1(f), 3(d)** and **4(c)**.

1 A file requires 2 gigabytes of storage.

(a) How long will it take to download this file if the download speed is 64 megabytes per second? You must show all your working.

_____ seconds [4]

(b) Convert the decimal number **–67** to a two's complement binary number using 8 bits. You must show all your working.

[3]

(c) Convert the two's complement binary number $\mathbf{01111111_2}$ to decimal. You must show all your working.

[2]

(d) How many positive decimal numbers can be represented using an 8 bit two's complement system?

_____ [1]

(e) How many negative decimal numbers can be represented using an 8 bit two's complement system?

_____ [1]

[Turn over

(f) In the past, character sets were restricted to 128 characters.

However, computers can now represent many more characters.

Explain and evaluate this development.

Quality of written communication will be assessed in your answer.

[6]

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(Questions continue overleaf)

[Turn over

2 The table below shows some annual sales for a shop.

Product	Actual Sales (£)	Target Sales (£)
Chocolate bars	32 661	35 000
Ice cream	18 974	20 000
Sweets	42 562	40 000

(a) By referring to the shaded cell in the table, explain what is meant by each of the following.

Data

[2]

Information

[2]

Knowledge

[3]

(b) Errors can occur during data entry.

(i) Explain how verification can be used to detect errors.

[4]

(ii) Explain what is meant by a transposition error.

[2]

[Turn over

(c) Describe each of the following methods of data validation.

Range check

Type check

Length check

[6]

(d) A chocolate bar has the product code 91167M where M represents the modulus 11 check digit.

Calculate the value for M, showing all your working.

M = _____ [5]

[Turn over

3 **(a)** Describe the role of each of the following during the fetch-execute cycle.

Program counter

Current instruction register

[4]

(b) Name and describe the three buses used during the fetch-execute cycle.

Name _____

Description _____

_____

Name _____

Description _____

_____

Name _____

Description _____

_____ [6]

[Turn over



(c) Explain how each of the following can influence the speed of a computer.

Processor type

Cache memory

[4]

(d) A computer can use an internal hard disk or a flash memory pen.

Compare the use of these devices for storage.

Quality of written communication will be assessed in your answer.

[6]

[Turn over

4 **(a)** Point of sale (POS) terminals are widely used in supermarkets.

(i) Identify four different input devices you would expect to find in a POS terminal.

1. _____

2. _____

3. _____

4. _____ [4]

(ii) Describe two ways in which a POS terminal will provide feedback.

1. _____

_____

_____

_____

2. _____

_____

_____

_____ [4]

(b) Explain why batch processing is suitable for a payroll application but unsuitable for a stock control application.

[4]

[Turn over

(c) Bitmap and JPEG are two file formats used for images. Evaluate the use of each of these formats for an image in a webpage.

Quality of written communication will be assessed in your answer.

[6]

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(Questions continue overleaf)

5 **(a)** Explain the following web technology terms.

URL

IP address

[4]

(b) Distinguish between the standards supported by the World Wide Web Consortium (W3C) and those supported by the International Telecommunications Union (ITU).

[6]

[Turn over

(c) Explain each of the following HTML tags.

<body>

<h1>

<p>

[6]

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(Questions continue overleaf)

[Turn over

(d) The following HTML code includes a cascading style sheet (CSS).

```
<html>
<head>
<style>

   body {background-color: lightblue;}

   h1 {font-family: arial;color: red;text-align: center;
   font-size: 40px;}

   p {font-family: verdana; color: green;text-align:
   center;font-size: 30px;}

</style>
</head>

<body>

<h1>Cascading Style Sheet</h1>

<p>CSS is a language that describes how the html web page
should be displayed</p>

</body>

</html>
```

Explain the purpose of the CSS used in this code.

[5]

THIS IS THE END OF THE QUESTION PAPER

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For Examiner's use only	
Question Number	**Marks**
1	
2	
3	
4	
5	
Total Marks	

Examiner Number

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